Kernix 0.2.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

07 workloads

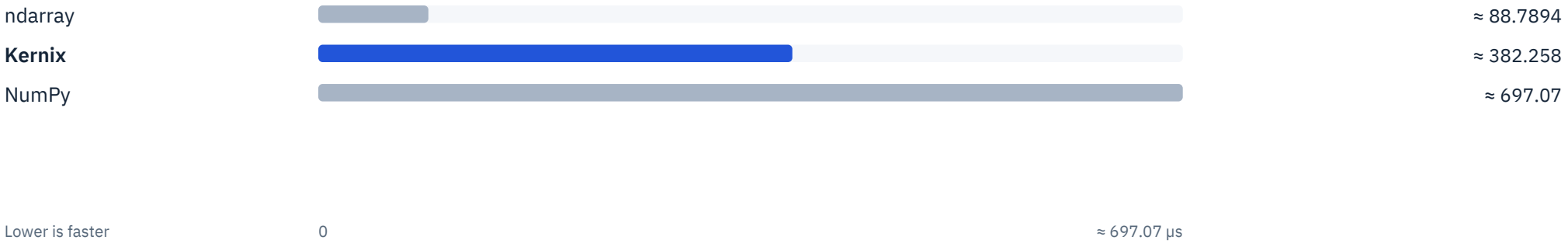
21 observations

Source precision retained

Matrix multiplication

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Historical medians of 15 batch means. Recorded p95 is the batch maximum.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

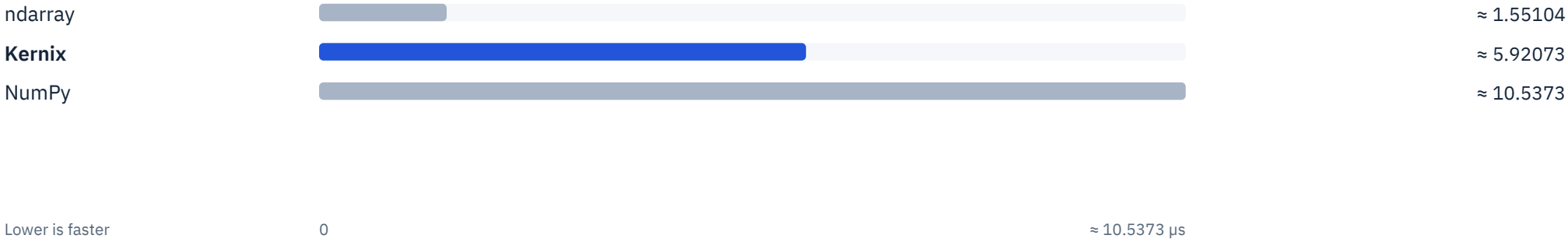
Kernix 0.2.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



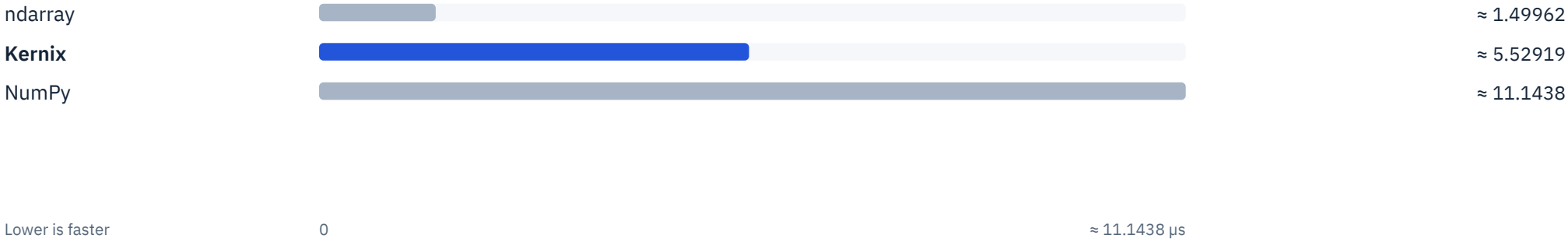
Kernix 0.2.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication · reused output

Batch-mean median

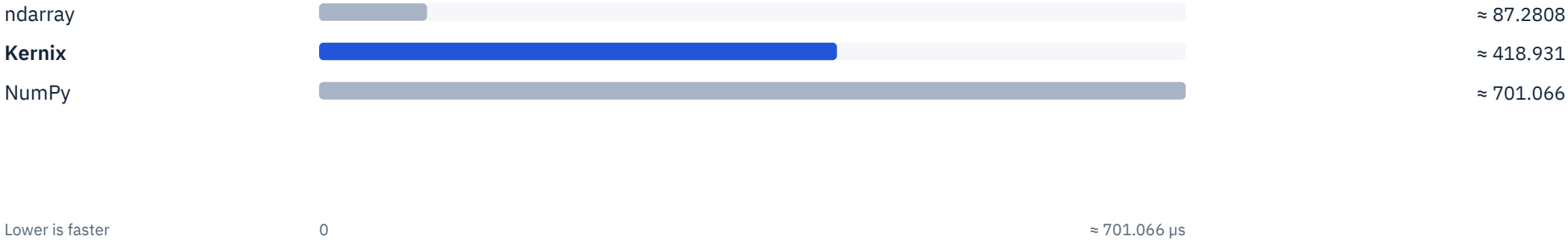
32 × 32 · 32 × 32 · f64 · 1 worker · reused output



Matrix multiplication · reused output

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · reused output



Kernix 0.2.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Elementwise addition

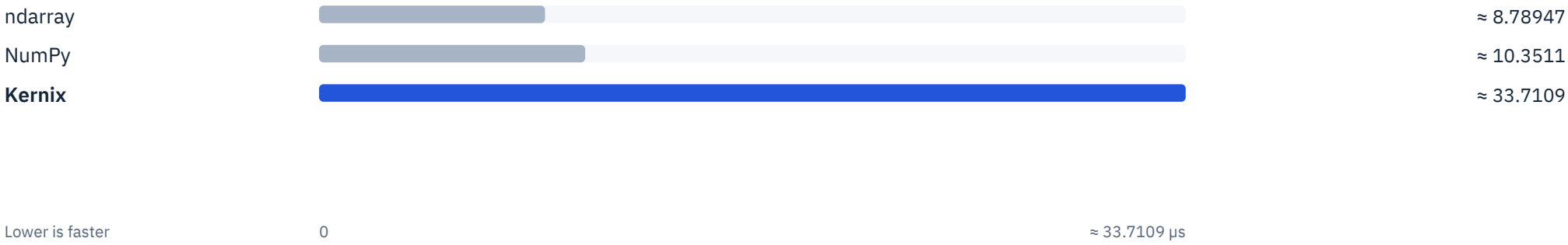
Batch-mean median

65,536 elements · f64 · 1 worker · allocated output

Sum reduction

Batch-mean median

65,536 elements · f64 · 1 worker · scalar output



Kernix 0.2.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Batch-mean median

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	5920.7331281079805	5.9207331281079805
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1551.0405732365532	1.5510405732365532
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	10537.275600505689	10.537275600505689
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	382257.98484848486	382.25798484848486
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	88789.4219858156	88.7894219858156
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	697070.3333333334	697.0703333333334
Matrix multiplication · reused output / Kernix 32 × 32 · 32 × 32 · f64	5529.18597965502	5.52918597965502
Matrix multiplication · reused output / ndarray 32 × 32 · 32 × 32 · f64	1499.6237178333633	1.4996237178333633
Matrix multiplication · reused output / NumPy 32 × 32 · 32 × 32 · f64	11143.829768270945	11.143829768270945
Matrix multiplication · reused output / Kernix 128 × 128 · 128 × 128 · f64	418930.75	418.93075

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · reused output / ndarray 128 × 128 · 128 × 128 · f64	87280.80139372822	87.28080139372822
Matrix multiplication · reused output / NumPy 128 × 128 · 128 × 128 · f64	701066.1111111111	701.0661111111111
Elementwise addition / Kernix 65,536 elements · f64	26130.575757575756	26.130575757575756
Elementwise addition / ndarray 65,536 elements · f64	24535.14019607843	24.53514019607843
Elementwise addition / NumPy 65,536 elements · f64	22793.959890610757	22.793959890610757
Sum reduction / Kernix 65,536 elements · f64	33710.90026954178	33.71090026954178
Sum reduction / ndarray 65,536 elements · f64	8789.467135325132	8.789467135325132
Sum reduction / NumPy 65,536 elements · f64	10351.081953642384	10.351081953642384
Row softmax / Kernix 256 × 64 · f64	221617.37168141594	221.61737168141594
Row softmax / ndarray 256 × 64 · f64	67250.7345844504	67.2507345844504

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / NumPy 256 × 64 · f64	92272.62361623616	92.27262361623616